Experiment 2:

Aim: To demonstrate a two-class confusion matrix using Python.

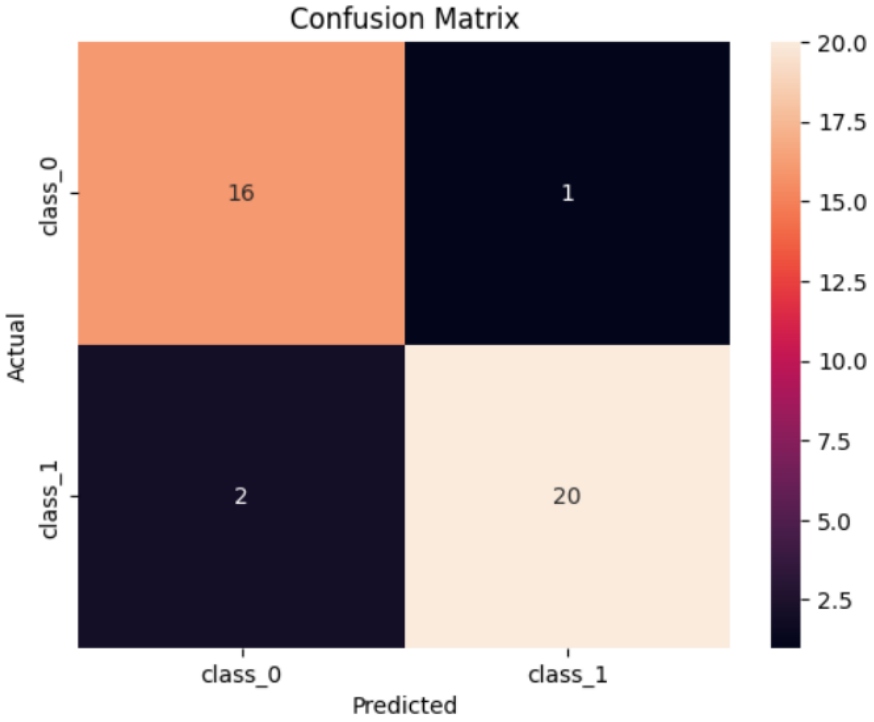
Code:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.datasets import load_wine
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score
from sklearn.metrics import confusion_matrix
wine = load_wine()
data = pd.DataFrame(wine.data, columns=wine.feature_names)
data['Target'] = wine.target
data = data[data['Target'] != 2]
X = data.drop('Target', axis=1)
y = data['Target']
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=1)
model = DecisionTreeClassifier(random_state=1)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
print("Accuracy:", accuracy_score(y_test, y_pred))
print("Precision:", precision_score(y_test, y_pred))
print("Recall:", recall_score(y_test, y_pred))
print("F1 Score:", f1_score(y_test, y_pred))
cm = confusion_matrix(y_test, y_pred)
sns.heatmap(cm, annot=True, fmt='d',
            xticklabels=wine.target_names[:2],
            yticklabels=wine.target_names[:2])
plt.xlabel("Predicted Label")
plt.ylabel("True Label")
plt.title("Confusion Matrix")
```

plt.show()

Output:

```
Accuracy: 0.9230769230769231
Precision: 0.9523809523809523
Recall: 0.9090909090909091
F1 Score: 0.9302325581395349
```



Result: Thus, the two-class confusion matrix was successfully implemented using a Decision Tree classifier in Python.